

magnetic fields

Important thing to note:

Electric fields are derived from arrangements of electric charge; magnetic fields come from **arrangements of magnetic poles**. Magnetic poles are always in pairs. Single magnetic monopoles have never been observed.

Magnetic field lines:

- Are in the direction in which a **north pole** would travel if free to do so.
- Cannot cross.
- Close lines $\implies$ strong field.
- Lines **begin on north poles** and end on south poles.
- Act as though they are elastic strings that repel each other but still try to be as short as possible. *I guess they want to be antisocial rubber midgets?*

